

3. Diketahui: $U = \{14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$

$A = \{14, 16, 18, 21, 24, 27, 29\}$; $B = \{17, 19, 21, 24, 26, 32\}$; $C = \{15, 21, 24, 25, 31, 34\}$

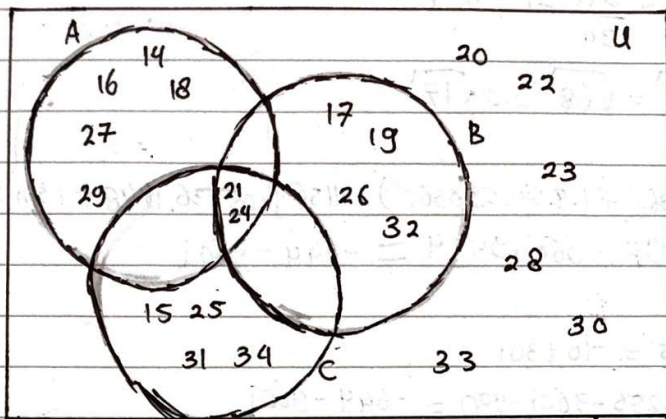
a. Diagram Venn

$(A \cup B \cup C) = \{14, 15, 16, 17, 18, 19, 21, 24, 25, 26, 27, 29, 31, 32, 34\}$; $(A \cap B \cap C) = \{21, 24\}$;

$(A \cup B \cap C) = \{14, 16, 17, 18, 19, 26, 27, 29, 32\}$;

* $A \cup B = \{14, 16, 17, 18, 19, 21, 24, 26, 27, 29, 32\}$; $C' = \{14, 16, 17, 18, 19, 20, 22, 23, 26, 27, 28, 29, 30, 32, 33\}$; $(A \cap B) = \{21, 24\}$

$(A \oplus B) = (A \cup B) - (A \cap B) = \{14, 16, 17, 18, 19, 26, 27, 29, 32\}$;



* Fungsi Eceran $f(x) = 6x + 5$; Fungsi subcharge $g(x) = \frac{x^2}{5} + 1$

$(f \circ g)(x) = f(g(x)) = 6\left(\frac{x^2}{5} + 1\right) + 5 = \frac{6x^2}{5} + 11$

$(g \circ f)(x) = g(f(x)) = \frac{(6x+5)^2}{5} + 1 = \frac{36x^2 + 60x + 25}{5} + 1 = \frac{36x^2}{5} + 12x + 6$

Apakah komposisi tsb komutatif? $\Rightarrow$ Komposisi tsb tidak komutatif, dapat dibuktikan dengan $x = b + 2 = 6$

$(f \circ g)(6) = \frac{6(6)^2}{5} + 11 = \frac{216}{5} + 11 = 54,2$; $(g \circ f)(6) = \frac{36(6)^2}{5} + 12(6) + 6 = \frac{1296}{5} + 72 + 6 = 337,2$

Berdasarkan pembuktian, dapat dinyatakan fungsi komposisi bersifat tidak komutatif ($54,2 \neq 337,2$)

d. $f^{-1}(x) \Rightarrow y = 6x + 5 \Rightarrow x = \frac{y-5}{6} \Rightarrow f^{-1}(x) = \frac{x-5}{6}$; Persamaan pendapatan total $\Rightarrow 6.100 + 144 = 605$ ribu

* Mencari volume $x \Rightarrow f^{-1}(605) = \frac{605-5}{6} = 100$; Jadi, volume pembelian yang menghasilkan Rp. 605.000 yaitu 100 unit.

e. Klasifikasi Fungsi Injektif, Surjektif, Bijektif

Fungsi $f(x) = 6x + 5$

Fungsi $g(x) = \frac{x^2}{5} + 1$

* Injektif ($x_1 = x_2$)

* Injektif X

$$6x_1 + 5 = 6x_2 + 5 \\ x_1 = x_2 \quad \checkmark$$

$$\frac{(x_1)^2}{5} + 1 = \frac{(x_2)^2}{5} + 1 \Rightarrow (x_1)^2 = (x_2)^2 \Rightarrow \pm x_1 \neq \pm x_2 \text{ (tidak sama, bisa saja } -x_1 = +x_2)$$

* Surjektif ($f(x) = y, y \in \mathbb{R}$)

* Surjektif (domain $= \mathbb{R} = x^2 \geq 0$) X

$$y - 5 = 6x \Rightarrow x = \frac{y-5}{6} \quad \checkmark$$

$$y = \left(\frac{x^2}{5}\right) + 1 \Rightarrow y - 1 = \frac{x^2}{5} \Rightarrow x^2 = 5(y-1) \Rightarrow 5(y-1) \geq 0 \Rightarrow y \geq 1$$

* Bijektif

$\hookrightarrow$ bukan surjektif, karena daerah hasil tidak hingga

Terbukti injektif dan surjektif,

* Bijektif

maka Bijektif $\checkmark$

Tidak terbukti injektif dan surjektif, maka bukan bijektif X